Sequential Conditional Means of Equicorrelated Scores

Including the fixed starting score in the average

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Each test has population mean 0 and standard deviation 1. Fix the first observed score at a. For jointly Gaussian tests with equal pairwise correlation c, recursively predict each subsequent test by conditioning on the starting score and on all earlier predictions. Every subsequent prediction equals ca. Including a as the first value, the average of $a, e_1, \dots, e_k$ is $a(1 + kc)/(k + 1)$. We derive the Gaussian conditional mean, solve the recursion, describe the admissible correlations and singular endpoints, and distinguish this deterministic construction from actual future scores. A counterexample shows that means, variances and correlations alone do not establish the result.

AI disclosure. This analysis, its mathematical exposition, supporting repository documentation and typesetting were prepared with OpenAI Codex and collaborating AI agents. The requester specified the problem, clarified the indexing and requested this analysis. Mathematical review and build checks were performed within the same AI-assisted workflow; no independent human peer review or new empirical psychometric validation is claimed.

1 Problem and notation

Let $k \geq 1$ be an integer and write

$$X_0, X_1, \dots, X_k$$

for standardized test scores. Throughout this analysis, every test has population mean zero and standard deviation one:

$$\mathrm{E}[X_j] = 0, \qquad \mathrm{SD}(X_j) = 1, \qquad \mathrm{Var}(X_j) = 1, \qquad 0 \leq j \leq k. \tag{1}$$

These parameters describe the reference distribution of each test; a fixed observed score a need not equal zero. We relabel the original tests $A_r(c, 0, 1)$ so that test 0 supplies the starting observation. For distinct indices, assume $\mathrm{Cov}(X_i, X_j) = \mathrm{Corr}(X_i, X_j) = c$.

Use $a \in \mathbb{R}$ for the *fixed starting score*; it is a scalar, not a function. For each $1 \leq n \leq k$, let g_n be the Gaussian conditional mean function of X_n given $(X_0, \dots, X_{n-1})$. Define

$$e_1 = g_1(a) = \mathrm{E}[X_1 \mid X_0 = a], \tag{2}$$

$$\begin{aligned} e_n &= g_n(a, e_1, \dots, e_{n-1}) \\ &= \mathrm{E}[X_n \mid X_0 = a, X_1 = e_1, \dots, X_{n-1} = e_{n-1}], \qquad 2 \leq n \leq k. \end{aligned} \tag{3}$$

The quantity of interest is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k + 1}.} \tag{4}$$

It contains $k+1$ values: the starting score a is the first value, followed by the k predictions $e_1, \dots, e_k$. Thus k counts subsequent predictions, while $k+1$ counts all values in the average.

For continuous distributions, conditioning on a particular score is not a ratio of probabilities of point events. A conditional mean is ordinarily defined only almost everywhere. Here, g_n means the continuous version supplied by Gaussian regression, derived below. This convention specifies the recursion even at individual histories of probability zero. The singular case $c = 1$ will be treated directly on its support.

2 The joint model and its correlation domain

Assume $(X_0, \dots, X_k)$ is jointly Gaussian, allowing degenerate Gaussian distributions at the endpoints. Its covariance matrix is

$$R_{k+1} = (1-c)I_{k+1} + c\mathbf{1}_{k+1}\mathbf{1}_{k+1}^{\mathsf{T}}, \tag{5}$$

where I_d is the d-dimensional identity and $\mathbf{1}_d$ is a column of d ones. Vectors orthogonal to $\mathbf{1}_{k+1}$ have eigenvalue $1-c$; the vector $\mathbf{1}_{k+1}$ has eigenvalue $1+kc$. Thus a covariance matrix for these $k+1$ scores exists exactly when

$$\boxed{-\frac{1}{k} \le c \le 1.} \tag{6}$$

An arbitrarily long equicorrelated collection requires $0 \le c \le 1$. A negative fixed c cannot satisfy the lower bound for all k.

For $0 \le c \le 1$, a construction consistent with the hypothetical common-factor illustration in the repository's Hoeflin/Ferguson analysis, Appendix A, is

$$X_j = \sqrt{c}\, G + \sqrt{1-c}\, \varepsilon_j, \tag{7}$$

where $G, \varepsilon_0, \dots, \varepsilon_k$ are independent standard-normal variables. The covariance-matrix derivation below also covers the admissible negative correlations. This is a mathematical model, not a claim that actual test batteries satisfy these assumptions.

3 Derivation of the conditional mean

Fix $1 \le n \le k$, initially with $c < 1$, and put

$$Y_n = (X_0, \dots, X_{n-1})^{\mathsf{T}}, \qquad R_n = \mathrm{Cov}(Y_n) = (1-c)I_n + c\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}.$$

The eigenvalue of R_n in the direction $\mathbf{1}_n$ is

$$d_n = 1 + (n-1)c.$$

It is positive throughout $-1/k \le c < 1$: for negative c, $d_n \ge 1 - (n-1)/k \ge 1/k > 0$. The other eigenvalues are $1-c > 0$. Consequently every conditioning block is nonsingular, even if the full $(k+1)$-score distribution is singular at $c = -1/k$.

An explicit inverse, verified by multiplying it by R_n, is

$$R_n^{-1} = \frac{1}{1-c}\left(I_n - \frac{c}{1+(n-1)c}\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}\right). \tag{8}$$

Since $\mathrm{Cov}(X_n, Y_n) = c\mathbf{1}_n^\mathsf{T}$, the regression coefficient is

$$\mathrm{Cov}(X_n, Y_n)R_n^{-1} = \frac{c}{1+(n-1)c}\mathbf{1}_n^\mathsf{T}. \tag{9}$$

To establish why this coefficient gives the conditional mean, define the residual

$$W_n = X_n - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} X_j. \tag{10}$$

For each $\ell \in \{0, \dots, n-1\}$,

$$\begin{aligned}\mathrm{Cov}(W_n, X_\ell) &= c - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1}\mathrm{Cov}(X_j, X_\ell)\\ &= c - \frac{c}{1+(n-1)c}[1+(n-1)c] = 0.\end{aligned}$$

The residual and Y_n are jointly Gaussian because they are linear transformations of a Gaussian vector. Their zero cross-covariance implies independence: the Gaussian characteristic function $\exp(-t^\mathsf{T}\Sigma t/2)$ factors across covariance blocks when the cross-covariance block is zero. This argument also allows a zero-variance residual. Since $\mathrm{E}[W_n] = 0$, independence gives $\mathrm{E}[W_n \mid Y_n] = 0$. Taking the conditional expectation in (10) therefore yields

$$\boxed{g_n(y_0, \cdots, y_{n-1}) = \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} y_j.} \tag{11}$$

This linear function is the unique continuous version, because the nonsingular vector Y_n has a strictly positive density everywhere.

For completeness, the same residual calculation gives

$$\begin{aligned}\mathrm{Var}(X_n \mid Y_n) = \mathrm{Var}(W_n) &= 1 - \frac{nc^2}{1+(n-1)c}\\ &= \frac{(1-c)(1+nc)}{1+(n-1)c}.\end{aligned} \tag{12}$$

This nonnegative conditional variance describes actual test outcomes, not randomness in the recursively specified numbers e_n.

4 Solving the recursion and taking the average

Inserting the prescribed history into (11) gives

$$e_n = \frac{c}{1+(n-1)c}\left(a + \sum_{j=1}^{n-1} e_j\right), \qquad 1 \le n \le k, \tag{13}$$

where the sum is empty at $n = 1$. In particular, $e_1 = ca$. Suppose $e_j = ca$ for all $1 \le j < n$. Then

$$e_n = \frac{c}{1+(n-1)c}[a + (n-1)ca] = \frac{ca[1+(n-1)c]}{1+(n-1)c} = ca.$$

Induction establishes

$$\boxed{e_1 = e_2 = \cdots = e_k = ca.} \tag{14}$$

Consequently the requested average is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k+1} = \frac{1+kc}{k+1}\, a = \left(c + \frac{1-c}{k+1}\right) a.} \tag{15}$$

Unlike the individual subsequent predictions, the requested average depends on the number of values included. With $\lambda_k = (1+kc)/(k+1)$, it is $B_k = \lambda_k a$. Once a, c, k are fixed, this average is deterministic. An additional expectation changes nothing: $\mathrm{E}[B_k] = \lambda_k a$. If $N = k+1$ denotes the *total* number of values, the same result is

$$\boxed{B_{N-1} = \frac{1+(N-1)c}{N}\, a, \qquad N \geq 2.} \tag{16}$$

For the single-value case $N = 1$, the average is simply a, without any subsequent prediction or inter-test correlation being needed.

There is also a partial-sum derivation. Define

$$T_0 = a, \qquad T_n = a + \sum_{j=1}^{n} e_j.$$

Equation (13) implies

$$T_n = \frac{1+nc}{1+(n-1)c} T_{n-1}.$$

Multiplying these relations telescopes, without dividing by c or by a:

$$T_n = a \prod_{j=1}^{n} \frac{1+jc}{1+(j-1)c} = a(1+nc).$$

Dividing $T_k = a + \sum_{j=1}^{k} e_j = a(1+kc)$ by $k+1$ again gives (15). This proof includes $a = 0$ and $c = 0$.

For example, if $a = 2$ and $c = 0.6$, then

$$e_1 = 1.2, \qquad e_2 = \frac{0.6}{1.6}(2 + 1.2) = 1.2, \qquad e_3 = \frac{0.6}{2.2}(2 + 1.2 + 1.2) = 1.2.$$

Every subsequent prediction remains 1.2. Including the starting score, the average after three predictions is instead

$$B_3 = \frac{2 + 1.2 + 1.2 + 1.2}{4} = \frac{1 + 3(0.6)}{4}\, 2 = 1.4.$$

5 Interpretation and other averages

5.1 Actual scores and their conditional average

The recursion replaces each previous future score by its conditional mean. Actual observed future scores need not equal those replacements. If

$$C_k = \frac{1}{k+1}\sum_{j=0}^{k} X_j$$

is the actual average including the starting test, bivariate Gaussian regression gives $\mathrm{E}[X_n \mid X_0 = a] = ca$ for every $1 \le n \le k$. Linearity therefore gives

$$\mathrm{E}[C_k \mid X_0 = a] = \frac{a + kca}{k+1} = \lambda_k a = B_k. \tag{17}$$

Although this conditional expectation equals B_k, the random variable C_k generally has positive conditional variance.

In fact,

$$\mathrm{Var}(C_k) = \frac{1+kc}{k+1} = \lambda_k, \qquad \mathrm{Cov}(C_k, X_0) = \lambda_k,$$

so Gaussian regression gives

$$\mathrm{Var}(C_k \mid X_0 = a) = \lambda_k - \lambda_k^2 = \frac{k(1-c)(1+kc)}{(k+1)^2}. \tag{18}$$

Likewise, the random predictors

$$P_n = \mathrm{E}[X_n \mid X_0, \dots, X_{n-1}]$$

use actual prior outcomes and are not the deterministic values e_n. The tower property implies $\mathrm{E}[P_n \mid X_0] = cX_0$, while its unconditional implication $\mathrm{E}[P_n] = 0$ holds without Gaussianity whenever the scores are integrable and have mean zero. Applying a conditional mean function at a fixed, recursively chosen history is a different operation.

5.2 The subsequent predictions considered separately

For comparison, averaging just the k subsequent predictions gives

$$\frac{1}{k}\sum_{n=1}^{k} e_n = ca. \tag{19}$$

The requested average B_k includes a as well and therefore has denominator $k+1$, as in (15).

5.3 Population mean zero and the fixed starting score

If one randomizes the starting observation according to its standardized test distribution, the recursively filled average becomes $\lambda_k X_0$. Consequently

$$\mathrm{E}[\lambda_k X_0] = 0, \qquad \mathrm{SD}(\lambda_k X_0) = \lambda_k, \tag{20}$$

where $\lambda_k \ge 0$ throughout the admissible correlation domain. These are population statements obtained by varying the starting score. For the fixed starting score in this problem, the answer is $B_k = \lambda_k a$, which need not be zero. Every original test still has mean 0 and standard deviation 1; the arithmetic average has its own distribution and is not rescaled to unit standard deviation.

6 Boundary cases

At $c = 0$, joint Gaussianity makes the scores independent, and every subsequent prediction is zero. Including the starting score gives $B_k = a/(k+1)$. At $c = 1$, $\mathrm{Var}(X_i - X_j) = 2(1-c) = 0$, so all scores agree almost surely. Given the starting score a, every compatible recursive prediction is $a = ca$, and $B_k = a$. This establishes the result at $c = 1$ without using the singular inverse in (8).

At the other endpoint, $c = -1/k$, the full vector satisfies

$$\sum_{j=0}^{k} X_j = 0 \quad \text{almost surely,}$$

because that sum has mean zero and variance $(k+1)(1+kc) = 0$. Every conditioning prefix remains nonsingular, as shown above, so the derivation of (15) applies. Its prescribed history is compatible with the support, since $a + kca = 0$, so $B_k = 0$. The final conditional variance (12) is zero.

7 Why correlation alone is insufficient

The original description specifies only marginal means, marginal variances and pairwise correlations. Those quantities do not determine the conditional mean functions. A counterexample already exists with one subsequent test. Let

$$P(X_0 = -\sqrt{2}) = P(X_0 = \sqrt{2}) = \tfrac{1}{4}, \qquad P(X_0 = 0) = \tfrac{1}{2}, \qquad X_1 = X_0^2 - 1.$$

Symmetry and direct calculation give

$$\mathrm{E}[X_0] = 0, \quad \mathrm{E}[X_0^2] = 1, \quad \mathrm{E}[X_0^3] = 0, \quad \mathrm{E}[X_0^4] = 2.$$

Consequently

$$\begin{aligned} \mathrm{E}[X_1] &= \mathrm{E}[X_0^2] - 1 = 0, \\ \mathrm{Var}(X_1) &= \mathrm{E}[(X_0^2 - 1)^2] = 2 - 2 + 1 = 1, \\ \mathrm{Cov}(X_0, X_1) &= \mathrm{E}[X_0^3 - X_0] = 0. \end{aligned}$$

Both scores are standardized and $c = 0$, yet at the fixed starting value $a = 0$, an event with positive probability,

$$e_1 = \mathrm{E}[X_1 \mid X_0 = 0] = -1 \neq 0 = ca.$$

The average including the starting value is therefore

$$B_1 = \frac{0 + (-1)}{2} = -\frac{1}{2},$$

whereas (15) would give zero under the Gaussian model. Therefore (15) is a result under the stated joint Gaussian model, or another model with the same conditional-mean functions; it does not follow from equicorrelation alone.

Scope and provenance. This derivation and its AI review extend the repository's hypothetical Gaussian model. Existing Ada/SPARK and Python/Deal evidence concerns the specified software; it does not formally verify this text. No new empirical study or formal-prover campaign was performed. The accompanying README records build and visual checks. The repository's original material follows its existing license, subject to third-party rights.